A Station-Level Air Quality Benchmark for Central Asia

Frozen leave-city-out splits, an operational-availability account, and an honestly-evaluated transfer baseline

> **Reproduction.** Every figure below is regenerated by `make reproduce` (or `python tasks.py reproduce`) from the frozen splits in `benchmark/splits/splits.json`, checksum `544a044c…f9e0c6c`. No number in this document is typed by hand.

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1. Introduction

1.1 The gap

Central Asia is among the most polluted inhabited regions on earth. It is also among the least instrumented. Tursumbayeva et al. (2023) put annual PM2.5 in six regional capitals at 4.3–12.6 times the WHO 2021 guideline and trace the burden mainly to coal combustion rather than transport, contradicting the official emissions inventories. The monitoring base under those numbers is thin and unevenly open. Turkmenistan runs no national air quality network at all. Kazakhstan releases its data only to users physically inside the country. Only Kyrgyzstan publishes in a fully open form (OpenAQ, 2025).

Estimates are not what the region lacks. Global gridded products already assign PM2.5 values across Central Asia, and the epidemiological literature consumes them. What nobody can do is check those estimates, or set two methods against each other on identical terms. There is no open station-level benchmark for the region. No frozen splits, no declared protocol, no shared evaluation.

The cost of that absence is measurable. Consider the closest environmental analogue in the literature: PM2.5 estimation across Xinjiang, an arid, dust-affected, sparsely monitored region much like this one. Jin et al. (2022) report R^2 between 0.73 and 0.81 under 10-fold cross-validation with no spatial stratification. With 41 stations in 16 cities and 8-day averaging, that design places observations from the same station, frequently the same window, on both sides of the split. The number answers one question. How well can we interpolate among stations we already have? Readers take it as answering a different one: how well can we estimate where no monitor exists? Nothing in a reported R^2 separates the two.

This is not a complaint about one paper. Roberts et al. (2016) show that spatially, temporally or hierarchically structured data require blocked validation, because ordinary k-fold leaves dependence straddling the split. Meyer et al. (2019) demonstrate the same for spatially derived predictors. Alazmi and Rakha (2022) measure the effect directly on particulate data, running random, spatial and spatiotemporal cross-validation side by side. Tang et al. (2024) place validation strategy among the systematically overlooked issues in air quality machine learning. The methodological position taken here is therefore not new. What the region has never had is a benchmark that enforces it.

1.2 What this paper contributes

AQ-Bench (Betancourt et al., 2021) is the precedent: 5,577 stations worldwide, split by spatial clustering at a 50 km threshold. It targets long-term ozone metrics from station metadata, a time-independent regression, and it excludes Central Asia entirely. We borrow its spatial clustering rationale for leave-station-out instead of inventing a second one, and diverge on pollutant, target, temporal protocol and region. Section 3 states the differences precisely. AirDelhi (Chauhan et al., 2023) offers a second precedent for fine-grained particulate benchmarking, confined to a single city.

C1 — a benchmark. 8 stations across 6 cities. Splits were frozen and hashed before a single model was fitted: blocked-temporal with a derived purge gap, leave-city-out over 6 folds, and leave-station-out where station density allows it (4 folds; Almaty, Ashgabat, Bishkek, Tashkent hold one instrument each and are named ineligible, not quietly dropped). A test enforces immutability. It fails for the authors exactly as it fails for anyone else.

C2 — transfer evaluated without flattery. We offer no new method. We measure how far spatial transfer survives into an aerosol regime unlike the source domain, under a protocol the region has never applied. That framing makes a negative result publishable. Several of ours are negative.

One fold deserves naming here. Khujand's stations begin after the training block closes, so the city contributes no training label anywhere in the record. Not one row. The model arrives with no local history of any kind and has to return a concentration regardless. This is the harshest test the benchmark contains, and it is also the one that matches the deployment case the work exists for: an unmonitored city asking for a number it has never been given. A model that collapses on Khujand has not earned the right to be deployed anywhere new. We report the fold on its own. Averaging it into the other five would report neither.

C3 — an operational-constraint account. Every predictor carries a measured latency and a typed availability flag. Anything that cannot exist at prediction time is barred from deployable configurations by test rather than by convention.

1.3 Findings that shaped the work

Three results recur, and each is worth stating before the methods.

A constant is hard to beat, and that says more about the region than about the models. No credential-free nowcaster beat a trivial always-exceed predictor on health-relevant exceedance. The reason is uncomfortable and entirely physical: PM2.5 in these cities clears the WHO 24-hour guideline on most days of the year, so a classifier that never varies is correct most of the time. Chronic pollution, not modelling skill, sets that floor. Beating it is the only evidence that a model has learned something past the regional mean. Raw CAMS, a full chemistry-transport model, is improved simply by subtracting a per-city constant, and even after that correction it reaches only R^2 -0.22. Across two phases, every baseline we tried sat below the accuracy of predicting the held-out city's own mean.

Metric choice reverses conclusions. RMSE and exceedance skill rank the same models in nearly opposite orders. Smoothing toward the mean lowers squared error while destroying the variance needed to tell a bad day from a good one, so a model wins one metric by losing the other. Report either alone and the ladder comes out backwards for the other task.

The first positive leave-city-out R^2 came from removing two handicaps, not from a better model. Tuned gradient boosting with spatial neighbour encodings reaches R^2 0.07 at RMSE 25.70, against 31.09 for bias-corrected CAMS, Diebold–Mariano -4.52 at $p < 0.0001$. The same architecture untuned and stripped of neighbours scored RMSE 31.99 and R^2 -0.64. Worse than the baseline it now beats.

1.4 What this paper does not claim

We do not claim state of the art. No like-for-like comparison exists on these splits, for the plain reason that we are the ones creating them. Nor do we claim to introduce transfer learning for data-poor regions. Gupta et al. (2024) hold that ground already, proposing a latent dependency factor for spatial transfer of PM2.5 estimation and reporting a 19.34% gain over baselines across ten target sensors against eastern-US source data. Theirs is a method paper, not a benchmark, and it defines no reusable public split. What remains available to us is the evaluation protocol and the region, not the idea. We also decline to compare our figures against published R^2 values obtained under random cross-validation. A leave-city-out number and a random-CV number are not commensurable, and treating them as though they were is the precise error this benchmark exists to prevent.

One piece of prior art we cannot resolve. A 2025 conference abstract claims the first machine-learning PM2.5 prediction for Tashkent, using ten automated stations with weather and seasonal inputs. We could not obtain it. The publisher returns HTTP 403, and the record appears in none of OpenAlex, Crossref, Semantic Scholar or Europe PMC. **Its split protocol is unverified.** We therefore make no claim about what that work did or did not do, and we specifically do not assert priority for applying leave-city-out to Tashkent. Should it prove to be spatially stratified, the novelty of C1 narrows to the multi-city benchmark, not the city. Section 7 carries this as an open item.

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2. Data, Operational Constraints, and Informative Missingness

2.1 Ground truth

Reference PM2.5 comes from 8 instruments across 6 cities: Almaty, Ashgabat, Bishkek, Dushanbe, Khujand and Tashkent. Six of the eight are US diplomatic-post reference monitors, which are the only consistent multi-country reference in the region and the sole route to any measurement in Turkmenistan.

That network is now closed. The US State Department ended its global embassy air quality programme in March 2025, and six of our eight stations terminate on exactly 2025-03-04. Reporting states that seventeen years of archive were subsequently removed from the originating platform. The record this benchmark curates is therefore finite, closed, and partially deleted at source — which raises rather than lowers the value of freezing it.

Quality control was pre-registered before the data were inspected, so no rule could be tuned to improve a count. Seven rules cover physical range, flatlining, zero-runs, unit sanity, duplicate identity, timezone correctness and completeness. Two produced findings worth reporting.

Duplicate identity. The embassy monitors are published twice — once by StateAir, once by AirNow — under distinct identifiers 57 m apart in Bishkek and 40 m in Ashgabat. Exact coordinate matching does not catch this. Under leave-station-out the held-out station would have been the same physical device as one in training. Detection is therefore distance-based at 150 m, with a wide margin: the two genuinely distinct Dushanbe sites are 6.06 km apart.

Timezone correctness. Metadata offsets are not trusted. Each station's diurnal composite is cross-correlated against a regional reference, and an initial implementation rejected both Khujand sensors for an apparent 12-hour shift. Investigation showed the regional reference self-correlates at r = +0.71 under a 12-hour rotation, because Central Asian urban PM2.5 is bimodal with peaks roughly half a day apart. Whole-shape correlation cannot separate a half-day offset from none in such a signal. The check now reports lag identifiability and flags rather than rejects when the hypothesis is not distinguishable — a station was nearly deleted by a discriminator reporting an artefact.

2.2 The splits

Train 2018-11-27 to 2022-12-31; validation 2023-01-11 to 2023-12-21; test 2024-01-01 to 2024-12-31. Purge gaps of 240 hours separate the blocks.

The purge is derived, not chosen. A training sample at *t* reads features over [*t*–max_lag, *t*] and predicts *t*+*h*, so the gap must satisfy purge ≥ max_lag + max_horizon = 168 + 72 = 240 hours. A test recomputes this from the model definitions, so introducing a model with a longer feature window fails the build rather than silently leaking across the boundary.

Test year 2024 is forced by the embassy shutdown. It is the last complete year with reference-grade coverage. A post-shutdown block would cover two cities, not six.

Khujand contributes no training rows at all. Both its stations begin in late 2023, after the training block closes, so Khujand appears only in validation and test. This passed the completeness rule because that rule checks span and coverage, not overlap with the training block. We retain it deliberately: it constitutes a pure zero-shot spatial transfer fold, in which the model has no local history in any form. Its fold is strictly harder than the others and is reported separately rather than averaged in.

2.3 Operational availability as a typed property

Reanalysis products describe exactly the quantities that drive PM2.5, are better documented than their forecast counterparts, and already sit in most atmospheric pipelines. A model consuming ERA5 boundary-layer height to predict tomorrow's air quality will post an excellent number and cannot be deployed. Nothing in its RMSE looks wrong.

Availability is therefore declared per feature and enforced by test. Every measured latency in this study is given in Table 2.1; **three of five initial estimates were wrong, one by more than three orders of magnitude.**

Product	Claimed	Measured	Status
MAIAC AOD (Earth Engine, MCD19A2.061)	6 h	~8 days	oracle only
Sentinel-5P AAI (OFFL)	5 h	~72 h	oracle only
Sentinel-5P AAI (NRTI)	—	< 24 h	deployable
CAMS forecast PM2.5	12 h	~10 h	**deployable**
ERA5 reanalysis	~5 days	163 h	oracle only
VIIRS active fire (v001)	4 h	**774 days**	superseded

The VIIRS case is the instructive one, and the latency was the lesser problem. The originally-mapped collection is deprecated and its final asset falls on 2024-06-16 — dead centre of the frozen test year, giving 161 images in January–June and **zero** in July–December. A model using it would have seen fire signal for half the test period and structurally none for the other half, producing an apparent regime change on 1 July that invites a meteorological explanation for a dead data feed. A latency check catches the 774 days; only a coverage check against the *frozen* test block catches the mid-block termination. Both are now enforced.

CAMS requires a second operational distinction: forecast step zero has assimilated observations at the valid time, so using it to predict that time is lookahead wearing a forecast label. All CAMS features use the 24-hour lead, which is what a live service holds.

2.4 The R7 phenomenon: missingness correlated with the target

Satellite retrieval does not fail at random. It fails during dust, cloud and snow — the conditions that accompany the highest concentrations. Dropping incomplete rows therefore conditions the evaluation on *"retrieval succeeded"* and biases every result toward calm, clear, low-concentration days.

We quantified this across all five satellite features on daily station-matched data. Each satellite product is extracted onto a complete station × date grid — the compositor emits a fully-masked image on days with no usable granule rather than omitting the row — so a failed retrieval is a masked cell, not an absent one, and the grid is the denominator. Rates below are conditional on the station-day also carrying a ground observation, since that is the subset on which the association can be tested at all.

Feature	Retrieval	Δ median PM2.5 on missing days	*p*	Retrieval on worst decile
Sentinel-5P SO_2	59.2%	**+10.6**	6.6e-128	**26.3%**
MAIAC AOD	64.6%	+4.8	3.8e-33	45.3%
Sentinel-5P NO_2	71.6%	+3.0	1.3e-12	60.0%
Sentinel-5P CO	82.7%	-0.4	0.24	85.8%
Sentinel-5P AAI	99.8%	+1.5	0.46	100.0%

SO_2 is blind in the season it exists to observe. It retrieves on 0.1% of December days against 93.5% in July. Retrieval requires ultraviolet signal, and at 38–43°N in December the solar zenith angle leaves too little; this is a geometric floor, not a cloud accident. The direct tracer for the region's dominant winter source is unavailable throughout that source's season. Additionally, 32.7% of the retrievals that do occur are negative, sitting below the noise floor — clipping them at zero would bias the coal tracer upward across a third of its observations.

The split between clean and contaminated features follows retrieval physics. CO uses the 2.3 µm shortwave-infrared band and AAI uses ultraviolet reflectance *ratios* rather than absorption depth; both survive winter geometry and cloud, and neither shows target-correlated missingness. MAIAC (visible/near-infrared) and the ultraviolet absorption retrievals do not.

The two clean features are complementary to the contaminated ones. AOD and AAI are both present on 64.5% of station-days; AAI alone covers a further 35.3%; neither is available on 0.1%. Usable satellite coverage therefore rises from 64.5% to 99.9%, and the coverage AAI adds is concentrated precisely where MAIAC fails.

Consequently missingness is **modelled, never dropped**. Valid-pixel counts are promoted to features in their own right. Section 6 shows they earn 4.1% of total SHAP attribution — comparable to the 5.1% contributed by the full chemistry-transport model — confirming that *when* a retrieval failed carries information about the atmosphere.

3. Benchmark Construction

The benchmark is the primary contribution. This section states what is frozen, what a submission must satisfy, and which failure modes are enforced mechanically rather than requested in prose.

3.1 What is frozen

`benchmark/splits/splits.json` fixes 8 stations across 6 cities, the three temporal blocks, 6 leave-city-out folds and 4 leave-station-out folds. It is accompanied by `splits.sha256`. The checksum is verified in CI and by `make reproduce`; any edit to the split file fails the build.

The split design follows the blocked-validation position of Roberts et al. (2016) for data with spatial and temporal structure, and adopts the spatial-clustering rationale of AQ-Bench (Betancourt et al., 2021) for leave-station-out. Leave-one-out protocols carry their own failure mode — Austin et al. (2025) show that distributional bias can compromise them — which is why leave-*city*-out, not leave-station-out, is the headline protocol: it holds out a whole aerosol regime rather than one instrument within a regime.

The checksum is over bytes, which on a mixed-platform repository is not automatic. An early version was unverifiable because `Path.write_text` applied CRLF translation on Windows, and the `.gitattributes` fix that followed was itself wrong: for attributes the **last** matching line wins, the opposite of `.gitignore`. The general `* text=auto` rule must therefore be declared *before* the `benchmark/splits/** -text` override, not after. This was caught only by cloning into a temporary directory and running `sha256sum -c` there — an in-place check passes on a file that no other machine can reproduce.

3.2 Two tasks, never pooled

	Task N — nowcasting	**Task F — forecasting**
Question	concentration at an **unmonitored** location, now	concentration at a **monitored** station, later
Protocol	leave-city-out	blocked temporal, t+24 h, t+48 h, t+72 h
Target history	**inadmissible**	admissible
Spatial features	admissible (neighbours exclude the held-out city)	admissible

The separation is not stylistic. Under leave-city-out the held-out city contributes no label, so a local autoregressive lag is undefined at inference — not merely optimistic, but unavailable. Under blocked-temporal forecasting the same lag is exactly what a deployed service holds. A single table containing both tasks would compare models with access to the target's own history against models without it, and the resulting ranking would be an artefact of that asymmetry. The two are reported separately throughout, and a test asserts that no results file mixes them.

3.3 The purge gap is derived

Purge = max_lag + max_horizon = 168 + 72 = 240 hours, applied at both block boundaries. A test recomputes this from the registered model definitions rather than reading a constant, so adding a model with a longer feature window fails the build instead of quietly leaking across the boundary. The failure mode this prevents is specific: a 168-hour rolling feature computed at the first test timestamp reads 168 hours of training labels.

3.4 Feature admissibility is typed, not documented

Every feature carries `available_at_runtime` and `latency_hours` as properties, and feature sets are constructed by filtering on them:

- `static_only` — geography and land surface; no temporal input at all
- `deployable` — everything a live service can hold at inference time
- `benchmark_retrospective` — adds products whose latency exceeds the forecast horizon
- `reanalysis_oracle` — deliberately inadmissible; exists to quantify the gap

The oracle set is retained precisely because it is not deployable. Reporting it alongside the deployable set converts an invisible methodological error into a measured quantity: the difference between the two is the cost of the operational constraint, and Section 6 reports it. A guard test constructs a contaminated feature set and asserts the admissibility check rejects it, so the check cannot silently degrade into a no-op.

3.5 Khujand: zero-shot spatial transfer

Both Khujand stations begin after the training block closes, so Khujand appears only in validation and test. Its leave-city-out fold is therefore strictly harder than the other five: the model has no local history in any form, not merely no label in the current fold.

We retain it as a distinct evaluation regime rather than repairing it. Every other fold measures interpolation between cities the model has seen at some point in training; Khujand measures extrapolation to a city that never appears. Averaging the two would report neither. Khujand is reported both inside the pooled figure and separately, and its fold is labelled zero-shot wherever it appears.

3.6 What a submission must report

1. Both tasks, in separate tables, with the ladder of Section 4 included.
2. Mean ± standard deviation over at least five seeds. Deterministic models report `(det.)` rather than `± 0.000`, so that a zero is never mistaken for a measured variance.

3. Per-city results, not only the pooled figure. Six cities with different aerosol regimes averaged into one number hides the variation that matters.

4. Diebold–Mariano tests (Diebold and Mariano, 1995) for any claimed improvement, with the truncation lag stated, Newey–West HAC variance (Newey and West, 1987) and the Harvey–Leybourne–Newbold small-sample correction (Harvey et al., 1997). Section 4.4 shows why the lag cannot be left at its default: setting it to zero inflated our *p*-values by seven orders of magnitude.

5. Exceedance metrics accompanied by the trivial-classifier score and Peirce skill (Section 4.3), because F1 alone does not establish skill.

6. A statement of which feature set was used, from the four above.

3.7 Reproduction

`make reproduce` runs the pipeline end to end: lint, type check, the full test suite, checksum verification, split regeneration, the baseline ladder, table regeneration and manuscript rendering — in that order, because splits are frozen and hash-verified before any model sees data. Every number in this paper is produced by that command. No figure in the manuscript is typed by hand; Section 7.5 describes the mechanism and the drift it caught. The command is idempotent: a second run leaves the working tree unchanged.

What that command does and does not require. The frozen splits are committed, so *verifying* the benchmark needs neither credentials nor a rebuild — `sha256sum -c splits.sha256` is sufficient, and the test suite runs offline against committed fixtures. *Regenerating* the numbers additionally requires the derived ground-truth panel, which is built from the OpenAQ archive with an API key. That panel is not redistributed here because the per-station licence terms are not yet transcribed; `data/MANIFEST.md` records the provenance and `README.md` the two commands that rebuild it. Running the pipeline without it fails with those instructions rather than a missing-file traceback. We state this explicitly because "one command reproduces everything" is a claim reviewers test, and it is true only after that acquisition step.

Two properties of that command are load-bearing and were not true of earlier versions. First, it **regenerates the manuscript, not only the tables**: an earlier `paper` target stopped after building the CSVs, so the rendered sections were never rebuilt from the figures `reproduce` had just recomputed, and the zero-drift guarantee held only halfway. Second, a missing tool is a **failure, not a skip** — the task runner once printed `SKIP (not installed)` and continued, so on a machine without `ruff` the command exited 0 having verified nothing. A reproduction harness that passes without doing the work is worse than no harness, because it converts an unrun check into a green light.

Environment setup is `python scripts/setup_env.py`, which works with or without `uv` and locates a Python 3.12 by *executing* candidate interpreters rather than trusting a registry. On our own Windows machine the `py` launcher advertised a 3.12 runtime that failed to start. Where `make` is unavailable, `python tasks.py <target>` runs the identical commands — the

Makefile delegates to the same table, so the two cannot diverge.

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4. The Baseline Ladder

A learned model is only interesting relative to what it beats. This section reports the mandatory ladder — persistence, climatology, raw chemistry-transport, and the spatial interpolators — before any learned model appears. Every rung is credential-free and deterministic, so seed variance is zero by construction and is reported as `(det.)` rather than as a measured `± 0.000`.

4.1 Task F — forecasting at monitored stations

Test block 2024, horizons t+24 h, t+48 h, t+72 h, pooled across stations.

Model	RMSE (μg/m³)	R^2
persistence, y(t)	42.78	-0.30
diurnal persistence, y(t–24 h)	42.78	-0.30
climatology (station mean)	38.15	-0.09
same-hour 7-day mean	35.58	0.13

Persistence and diurnal persistence are numerically identical, and this is correct. All three evaluated horizons are multiples of 24 h, so "the same hour yesterday" and "lag-24 persistence" resolve to the same timestamp. The Diebold–Mariano procedure reports them as exact ties — the loss differential is identically zero and the test statistic is undefined rather than insignificant. We report the degeneracy rather than dropping a rung, because a ladder in which two rungs coincide at the evaluated horizons is a property of the horizon grid that a reader needs in order to interpret the table.

Degradation with lead time is visible only in the persistence family (40.27 → 44.26 μg/m³ from t+24 h to t+72 h). Climatology is flat by construction at 38.15 μg/m³, since it does not read the recent past. **Only the same-hour 7-day mean achieves positive R^2** (0.13); every other rung is worse than predicting the test-block mean.

4.2 Task N — nowcasting under leave-city-out

The held-out city contributes no label. These are pure spatial interpolators, fitted on the remaining five cities.

Model	RMSE (μg/m³)	R^2	Exceedance F1	Peirce skill
nearest monitor	48.03	-0.71	0.737	0.414

IDW (k=5, p=2)	43.65	-0.32	0.762	0.317
training-pool mean	43.09	-0.27	0.764	0.000
ordinary kriging	40.92	-0.14	0.728	0.295

Every spatial baseline has negative R^2. Interpolating between Central Asian cities hundreds of kilometres apart is worse than predicting a constant. This is the honest floor the task sits on, and it is the reason the leave-city-out result in Section 6 is modest rather than impressive.

The training-pool mean is an explicit rung, not an afterthought. It is a constant: the mean of all training-block labels. It was promoted to the ladder after we observed that it outranks two of the three genuine interpolators on RMSE (43.09 vs 43.65 and 48.03). Any model that does not clear a constant has not demonstrated spatial skill, and without this rung in the table the reader cannot check that.

Kriging attains the best RMSE (40.92) and the **worst** exceedance F1 (0.728). Optimising squared error pulls predictions toward the mean, which suppresses exactly the excursions the exceedance metric scores. Reporting one number without the other would support two opposite conclusions about the same model. We additionally report the kriging fallback rate: where the system is singular the implementation silently degrades to IDW, and a column labelled "kriging" that is substantially IDW underneath is a reporting error rather than a modelling one.

4.3 Exceedance F1 does not measure skill

The WHO 2021 24-hour guideline is 15 µg/m³, scored on local-calendar daily means because that is what the guideline defines. In this region the exceedance base rate is **64.8%** — exceedance is the common case, not the rare one.

A classifier that predicts "exceeds" unconditionally therefore scores F1 = **0.764**. The training-pool mean — a constant, carrying no information whatsoever — scores 0.764, and is the **highest-F1 model in the entire Task N ladder**. Its Peirce skill is 0.000, exactly zero, as it must be for any constant.

This is not a curiosity about one table. It means a paper reporting only exceedance F1 on this region could present a constant as its best classifier and the number would look respectable. We therefore report, alongside every exceedance F1:

- `f1_trivial_always` — the score of the unconditional classifier on the same rows
- `beats_trivial` — whether the model exceeds it at all
- `base_rate` — so the reader can compute the trivial score independently
- `peirce_skill` — TPR – FPR, which is base-rate independent and **zero for any constant**

Peirce skill separates the ladder in the way F1 does not: nearest monitor (0.414) and IDW (0.317) carry genuine discriminative information, while the constant carries none, and the F1 column ranks them in the opposite order.

4.4 CAMS as a baseline

CAMS is the operational chemistry-transport forecast for the region and the strongest credential-free comparator. It is used at the 24-hour lead throughout: forecast step zero has assimilated observations at the valid time, so scoring it against that time is lookahead wearing a forecast label.

Variant	RMSE (μg/m³)	R^2	Bias (μg/m³)
raw	37.21	-0.42	-22.79
locally debiased	31.10	-0.12	-0.38
pooled debiased	32.57	-0.22	-0.38

Raw CAMS under-predicts by -22.79 μg/m³ — a large systematic deficit consistent with an emissions inventory that does not capture residential coal and biomass combustion at Central Asian intensity. Removing that bias improves RMSE substantially (37.21 → 31.10) while R^2 remains negative (-0.12): the correction fixes the level, not the timing.

The two debiasing variants are distinguished by protocol. **Local debiasing is admissible only in Task F**, since it fits a per-city offset on that city's training labels. Under leave-city-out no such labels exist, so Task N uses the pooled correction fitted with the held-out city excluded. The pooled variant is the weaker of the two (32.57 vs 31.10), and reporting the local figure under a leave-city-out heading would overstate the baseline the learned model must beat — making the model's margin in Section 6 look smaller than it is, in this direction, but the protocol violation is the same either way. A test asserts that a bias fitted on the full record differs from one fitted on the training block, so that a silently-leaking implementation cannot pass.

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5. Models and Training Protocol

5.1 Model class

Gradient-boosted decision trees (LightGBM) are the learned model. The choice is deliberate and is not a claim that trees are the strongest possible architecture. Three properties matter for this benchmark:

1. **Native missing-value handling.** Section 2 established that satellite retrieval failure is correlated with the target. A model that requires imputation forces a choice between discarding the informative rows and inventing values for them; LightGBM learns a default

direction per split, so missingness is used rather than filled.

2. **Sample efficiency.** Six cities is a small spatial sample. A deep architecture would be tuned on validation folds of a few hundred station-days, and the tuning variance would exceed the effect being measured.

3. **Attribution that survives inspection.** Section 6 depends on being able to say which feature families carry the model, and to show that the answer is uncomfortable.

Reporting a deep model here would require a like-for-like comparison we cannot yet support at this sample size; the benchmark exists so that such a comparison can be made later on fixed splits.

5.2 Feature families

- **spatial neighbour** — inverse-distance-weighted neighbour concentration, neighbour mean, neighbour count. Under leave-city-out the neighbour set **excludes the held-out city**, which is enforced by passing the held-out label into the constructor rather than by filtering afterwards.
- **static geography** — elevation, terrain basin indices at 6-city scale, population density, distance to the Aralkum.
- **calendar** — cyclical day-of-year and hour encodings, weekday.
- **satellite** — MAIAC AOD, S5P AAI/NO_2/SO_2/CO.
- **satellite missingness** — valid-pixel counts and retrieval indicators, promoted to features in their own right.
- **CAMS forecast** — the 24-hour-lead chemistry-transport prediction.

Autoregressive lags of the target are **Task F only**. They are not merely optimistic under leave-city-out; they are undefined, because the held-out city supplies no history. For Task N the spatial features above are the only route to target information, and they are constructed with the held-out city removed.

5.3 Tuning

Hyperparameters are selected on the 2023-01-11 to 2023-12-21 validation block and then frozen. The test block 2024-01-01 to 2024-12-31 is read exactly once per reported configuration. No hyperparameter, feature-set choice, or early-stopping decision is made against test-block performance.

Tuning was not cosmetic. Untuned defaults produce a Task N leave-city-out RMSE of 31.99 µg/m³ with $R^2 = -0.64$ — worse than the training-pool-mean constant of Section 4. Tuned, the same feature set reaches 25.70 µg/m³ at $R^2 = 0.07$.

Feature set	Untuned RMSE	Tuned RMSE	Untuned R^2	Tuned R^2
static only	33.22	28.31	-1.03	-0.24

deployable	31.15	25.75	-0.57	0.04
retrospective	31.99	25.70	-0.64	0.07

An untuned GBDT would have supported the conclusion that gradient boosting cannot beat a constant on this task. That conclusion would have been an artefact of the defaults, and it is the reason the untuned column is retained in the paper rather than discarded once better numbers existed.

5.4 Seeds and variance

Every configuration is run with five seeds and reported as mean ± standard deviation. Seed variance is small: the maximum across configurations is 0.24 µg/m³ for Task F and 0.85 µg/m³ for Task N.

Fold-to-fold variance is an order of magnitude larger than seed variance. The standard deviation of Task N RMSE across the 6 leave-city-out folds is 7.74 µg/m³, against 0.85 µg/m³ across seeds. Which city is held out dominates which seed was drawn. A study reporting seed error bars alone would communicate a precision this benchmark does not have, and per-city results are therefore mandatory (Section 3.6).

5.5 Operational cost of deployability

The deployable feature set excludes every product whose latency exceeds the forecast horizon; the retrospective set adds them back. The difference is the price of being deployable:

Task	Deployable	Retrospective	Cost
N (leave-city-out)	25.75	25.70	small
F (forecasting)	21.48	20.94	small

The cost is negligible in both tasks — 25.75 against 25.70 µg/m³ under leave-city-out, well inside the fold-to-fold spread of 7.74 µg/m³. This is a positive result for deployment and a deflationary one for the satellite features: products that arrive too late to be operational are also contributing little when they are available. Section 6.4 shows why.

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6. Results

All figures are on the frozen test block 2024-01-01 to 2024-12-31, five seeds, with Task N and Task F reported separately throughout.

6.1 Task N — leave-city-out nowcasting

Model	RMSE (µg/m³)	MAE	R^2
best spatial baseline (kriging)	40.92	—	-0.14
training-pool mean (constant)	43.09	—	-0.27
CAMS, pooled debias	31.09	23.29	-0.30
LightGBM, static only	28.31	19.22	-0.24
LightGBM, deployable	25.75	17.25	0.04
LightGBM, retrospective	25.70	17.38	0.07

The learned model beats every baseline, and its R^2 is 0.07. Both statements are load-bearing. It improves on pooled-debiased CAMS by 31.09 → 25.70 µg/m³ and on the best spatial interpolator by a wider margin. It also explains almost none of the variance in a never-seen city. We report this as the result: predicting urban PM2.5 at a location with no local monitor, from a training set of five other cities, is close to unsolved in this region, and the benchmark exists to make that measurable rather than to conceal it.

6.2 Per-city results and the Diebold–Mariano tests

Pooling six cities into one number hides the finding.

Held-out city	LightGBM RMSE	CAMS RMSE	DM statistic	*p*
Almaty	15.73	25.69	-5.48	0.0000
Ashgabat	21.34	20.75	0.35	0.7271
Bishkek	20.92	23.64	-1.87	0.0630
Dushanbe	38.58	48.81	-2.82	0.0050
Khujand (zero-shot)	31.55	39.08	-2.51	0.0124
Tashkent	26.35	28.56	-1.57	0.1180
pooled (n = 2480)	**28.97**	**35.68**	**-4.52**	**< 0.0001**

Tests use Newey–West HAC variance with the Harvey–Leybourne–Newbold small-sample correction. Negative statistics favour the learned model.

The pooled improvement is significant (< 0.0001); only 3 of 6 individual folds are. Bishkek (0.0630) and Tashkent (0.1180) show lower RMSE that does not clear significance, and we do not describe those as improvements. **Ashgabat favours CAMS outright** (21.34 vs 20.75 µg/m³, DM statistic 0.35, p = 0.7271) — the sign is reversed and the result is far from significant, so the honest summary is that the two are indistinguishable there rather than that CAMS wins.

Khujand, the zero-shot fold, is significant (0.0124) at 31.55 µg/m³ against CAMS at 39.08. A city with no training rows at all is predicted better than by the operational chemistry-transport model. It remains among the hardest folds in absolute terms, second only to Dushanbe.

Truncation-lag sensitivity. The DM truncation lag is derived from the forecast horizon, and an early implementation passed `horizon_hours = 1` for the nowcasting task, which disables the HAC correction entirely and inflated *p*-values by roughly seven orders of magnitude. We now report the full sensitivity sweep: across truncation lags of 0 to 60 hours the pooled conclusion is stable, with all comparisons significant (yes) and a worst-case *p* of 0.0019.

6.3 Task F — forecasting at monitored stations

Feature set	RMSE (µg/m³)	R^2
best Task F baseline (same-hour 7-day mean)	35.58	0.13
LightGBM, static only	23.08	0.57
LightGBM, deployable	21.48	0.63
LightGBM, retrospective	20.94	0.65

Task F is the easier problem and the numbers say so: $R^2 = 0.65$ against 0.07 for Task N. **These two figures must never be quoted together as though they described one system.** The Task F model may read the station's own recent history; the Task N model is predicting a city it has never seen. Reporting 20.94 µg/m³ as "the model's accuracy" would describe a capability the deployment does not have at unmonitored locations, which is the case the artefact exists to serve.

6.4 SHAP attribution: what actually carries the model

Mean absolute SHAP over the test block, by feature family:

Family	Share of total attribution
spatial neighbour	**32.5%**
static geography	25.1%
calendar	16.8%
satellite	16.6%
CAMS forecast	5.1%
satellite missingness	4.1%

Spatial interpolation drives the model, not the satellite record. The single largest feature is `nbr_idw` at 11.54 mean absolute SHAP — inverse-distance weighted neighbour concentration — more than twice the second-ranked feature (`doy_cos`, 4.84). Spatial neighbours and static geography together account for 32.5% + 25.1% of attribution, while the five satellite products contribute 16.6% between them.

This is the paper's least comfortable result and we state it plainly: **the model is largely a well-tuned spatial interpolator with geographic priors.** The satellite features are not inert — 16.6% is not nothing, and Section 5.5 shows the deployable set loses little — but a reader would be entitled to expect, from a study built on five remote-sensing products, that those products carried the prediction. They do not.

Two further readings follow. First, **satellite missingness earns 4.1% — comparable to the 5.1% contributed by the entire chemistry-transport model.** Whether a retrieval failed carries nearly as much information as what the atmospheric model predicted, which vindicates modelling missingness rather than dropping it (Section 2.4) and simultaneously indicates how little the retrieved values themselves add. Second, calendar features at 16.8% confirm that a large part of the signal is seasonal regularity that any climatology captures — consistent with the same-hour 7-day mean being the only Task F baseline with positive R^2.

6.5 Summary of what is and is not established

Established:

- The learned model beats pooled-debiased CAMS across the 6-fold leave-city-out protocol, pooled p = < 0.0001.
- Deployability costs almost nothing: 25.75 vs 25.70 µg/m³.
- Zero-shot transfer to a city with no training rows beats CAMS significantly (0.0124).

Not established:

- Per-city superiority. 3 of 6 folds reach significance; Ashgabat does not favour the model at all.
- Useful absolute accuracy at unmonitored locations. $R^2 = 0.07$.
- That satellite remote sensing drives the result. Section 6.4 shows it does not.

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7. Limitations and Discussion

This section is written for a reader hunting the study's weakest point. We would rather state it than have it found.

7.1 The labels themselves are provider-dependent in one city

The embassy monitors are published twice, by StateAir and by AirNow, under distinct identifiers. Ashgabat's pair is a clean duplicate: the two feeds agree to 0.1 μg/m³ on 100.0% of overlapping test-block hours. **Bishkek's pair is not.**

Across the full record the Bishkek feeds agree on 52.0% of overlapping hours, and the divergence concentrates in the frozen test year. Over 5389.00 overlapping hours in 2024 they agree on only **11.1%**, with a 95th-percentile disagreement of **33.6 μg/m³**. That is comparable to the RMSE of every model in this paper.

The implication is uncomfortable and irreducible. For Bishkek in the test block there is no single ground truth. Choosing the other provider's feed would shift the reported error by roughly the margin that separates our models from each other. Bishkek's DM result (p = 0.0630, not significant) should be read with that in view, and we claim no improvement there. Merging the feeds, which we do, reduces variance. It cannot manufacture a label the two publishers agree on. **This limits the data source, not the pipeline, and no modelling choice removes it.**

7.2 Leave-station-out covers a minority of the benchmark

4 leave-station-out folds exist and they span two cities. **Almaty, Ashgabat, Bishkek, Tashkent each hold a single instrument**, so within-city station holdout is undefined there.

The consequence runs deeper than reduced coverage. The Q6 timezone check compares instruments within a city. In a single-station city, a constant lifelong offset is invisible to every rule in the suite, and the QC output records that explicitly instead of returning a pass that actually means "not tested". Four of six cities therefore rest on metadata correctness for their time alignment. We regard this as the benchmark's largest unaudited assumption.

Leave-city-out stays the primary spatial protocol for one reason: it is available for all 6 cities. Leave-station-out is reported where possible and supports no headline claim.

7.3 The model does not beat CAMS everywhere

3 of 6 leave-city-out folds reach significance. In **Ashgabat the sign is reversed**: LightGBM 21.34 μg/m³ against CAMS 20.75 μg/m³, DM statistic 0.35, p = 0.7271. Read correctly, the two are indistinguishable there. CAMS does not win. The honest summary of the fold set is that the pooled result (< 0.0001) rests on Almaty, Dushanbe and Khujand, while Bishkek (0.0630) and Tashkent (0.1180) post lower RMSE that never clears significance.

Khujand carrying part of the pooled result is worth pausing on, since it is the fold with no training label at all. Zero-shot transfer into an unmonitored city is not where a reader would expect the method to hold up best. It does, and we take that as the strongest evidence in the paper that the spatial machinery generalises instead of memorising.

Ashgabat is the opposite case, and the one where the model has least to work with. Turkmenistan operates no national monitoring network, so the fold reduces to a single embassy instrument with no domestic context, and its nearest benchmark neighbour sits far outside any plausible interpolation radius. Under exactly those conditions a chemistry-transport model with a physical emissions inventory is a strong comparator. It should be.

7.4 The satellite record is not what carries the model

Section 6.4 is the result we would most like to have come out differently. Spatial neighbour features account for 32.5% of attribution and static geography a further 25.1%. The five satellite products together account for 16.6%. The top single feature is `nbr_idw`, inverse-distance-weighted neighbour concentration, at more than twice the second-ranked feature.

A study assembled around five remote-sensing products turns out, on inspection, to be largely a well-tuned spatial interpolator with geographic priors. Three qualifications follow, none of which overturn that.

1. *This is a property of the protocol as much as of the products.* Leave-city-out asks for a concentration where no monitor exists. Neighbour information is the most direct route to that answer, and satellite columns are a weak proxy for surface concentration under any protocol whatsoever.
2. *Missingness outperforms the values.* Satellite missingness earns 4.1%, comparable to the 5.1% contributed by the entire chemistry-transport forecast. *Whether* a retrieval failed is nearly as informative as what CAMS predicted. That is a finding about retrieval physics, and at the same time a measure of how little the retrieved values contribute.
3. *SO_2 is structurally absent in the season it exists to observe.* Retrieval needs ultraviolet signal; at these latitudes in December it falls to 0.1% against 93.5% in July. The direct tracer for the region's dominant winter source is unavailable throughout that source's season, so its low attribution is partly a measurement-geometry artefact, not evidence that SO_2 carries no information.

We report the attribution as measured. A paper arguing that satellite remote sensing enables air quality prediction in Central Asia would not be supported by these experiments.

7.5 Zero-drift reporting, and the drift it caught

Every number in this manuscript is a double-brace placeholder token resolved at render time from `paper/tables/*.csv`. An unresolved placeholder is a hard build failure. A test also checks a sample of manuscript figures directly against the CSVs, bypassing the intermediate `numbers.json` entirely, so that a bug in the extractor cannot certify itself.

The mechanism earned its place during drafting. Section 2's missingness statistics were originally transcribed from a console output. When they were finally banked to a table, the transcribed values proved wrong: SO_2 retrieval had been written as 61.5% against a recomputed 59.2%, and four further figures were off in the same direction. The errors were small and none of them changed a conclusion, which is exactly why no reader would ever have caught them. Regenerating the table also exposed that the two Section 2 tables had no producer script at all and could not be rebuilt by `make reproduce`. Both have one now.

7.6 Scope of the record

- **The reference network is closed.** Six of eight stations end on 2025-03-04 with the termination of the US diplomatic-post monitoring programme. **No result in this paper speaks to current conditions**, and the benchmark cannot be extended forward from this source.
- **Kazakhstan contributes one city.** Astana failed the completeness rule at 42.8%. The largest country in the region is represented by Almaty alone.
- **Six cities is a small spatial sample.** Fold-to-fold standard deviation (7.74 μg/m³) exceeds seed standard deviation (0.85 μg/m³) by roughly an order of magnitude. Conclusions are far more sensitive to which cities are in the set than to any training randomness.
- **ERA5 is oracle-only and incompletely retrieved.** Its measured latency (163 h) exceeds every evaluated horizon, so it cannot enter the deployable set. The multi-year retrieval was stopped once that was established.
- **One item of prior art is unresolved.** A 2025 conference abstract claims the first machine-learning PM2.5 prediction for Tashkent. The publisher page returns HTTP 403 and the record is absent from OpenAlex, Crossref, Semantic Scholar and Europe PMC, so **its split protocol could not be verified**. Section 1.4 states what we consequently do not claim. Should that work prove spatially stratified, the contribution of C1 narrows from "first such protocol in the region" to "first multi-city benchmark in the region". The benchmark is unaffected either way. Only the framing of its novelty moves.
- **Literature depth is uneven, and recorded as such.** Of 30 sources, 16 were read in full and 6 at abstract depth; 7 remain at snippet depth behind publisher paywalls. For those, bibliographic identity is verified against Crossref/OpenAlex but *content* claims are not independently confirmed. `research/LITERATURE.md` records the two axes separately rather than averaging them into a single confidence.
- **Russian-language coverage is incomplete.** CyberLeninka and eLIBRARY.RU are not indexed by the APIs used here, so the falsifier F4 verdict, that no equivalent regional benchmark exists in the Russian-language literature, remains provisional.

7.7 Train/serve skew is measured but not eliminated

The deployable set uses near-real-time satellite products while the benchmark is built on standard-latency ones. These are different processing streams over the same instrument, and we have not quantified the distributional difference between them. A service trained on standard products and served NRT products is exposed to skew that this study bounds only indirectly, through the small deployable/retrospective gap of Section 5.5. That gap is evidence that latency-restricted *feature sets* cost little. It is not evidence that NRT and standard versions of the same product are interchangeable. Quantifying the difference is the most immediate piece of follow-up work, and it needs a parallel NRT archive we did not have.

7.8 What the benchmark is for

The headline modelling result is modest. $R^2 = 0.07$ at unmonitored locations, a significant but not transformative improvement over CAMS, and an attribution profile dominated by spatial interpolation. We consider that the appropriate outcome.

The contribution is the fixed evaluation. Before this benchmark existed, a Central Asian air quality result could be reported on a random split, with reanalysis features unavailable at inference time, against no baseline ladder, scored with an exceedance F1 that a constant already achieves (0.764 at a 64.8% base rate, because the region's air is bad on most days, not because the classifier is good). Every one of those choices would have produced a more impressive paper than this one. The splits are frozen and checksummed. The protocol violations are enforced by failing tests rather than requested in prose. The numbers above are what survives that. A future model that genuinely improves on 25.70 µg/m³ under this protocol will have demonstrated something real.

References

Generated by `paper/scripts/stitch.py` from `research/sources.json`. Every entry was resolved through OpenAlex, Crossref, Semantic Scholar or Europe PMC and checked against a title-match gate; none is typed by hand.

- **[B3]** Asmaa Alazmi and Hesham Rakha (2022). *Assessing and Validating the Ability of Machine Learning to Handle Unrefined Particle Air Pollution Mobile Monitoring Data Randomly, Spatially, and Spatiotemporally*. International Journal of Environmental Research and Public Health. https://doi.org/10.3390/ijerph191610098
- **[B4]** George I. Austin, Itsik Pe’er and Tal Korem (2025). *Distributional bias compromises leave-one-out cross-validation*. Science Advances. https://doi.org/10.1126/sciadv.adx6976
- **[A7]** Jamie Banks, Bernd Heinold and Kerstin Schepanski (2022). *Impacts of the Desiccation of the Aral Sea on the Central Asian Dust Life■Cycle*. Journal of Geophysical Research Atmospheres. https://doi.org/10.1029/2022jd036618
- **[B1]** Clara Betancourt et al. (2021). *AQ-Bench: a benchmark dataset for machine learning on global air quality metrics*. Earth system science data. https://doi.org/10.5194/essd-13-3013-2021
- **[B5]** Sachin Chauhan et al. (2023). *AirDelhi: Fine-Grained Spatio-Temporal Particulate Matter Dataset From Delhi For ML based Modeling*. Neural Information Processing Systems. https://doi.org/10.52202/075280-3298

• **[A4]** Xiangyue Chen et al. (2021). *Validation and comparison of high-resolution MAIAC aerosol products over Central Asia*. Atmospheric Environment. https://doi.org/10.1016/j.atmosenv.2021.118273

• **[S:M3]** Francis X. Diebold and Roberto S. Mariano (1994). *Comparing Predictive Accuracy*. National Bureau of Economic Research. https://doi.org/10.3386/t0169

• **[C1]** Shrey Gupta et al. (2024). *Spatial Transfer Learning for Estimating PM$$_{2.5}$$ in Data-Poor Regions*. Lecture Notes in Computer Science. https://doi.org/10.1007/978-3-031-70378-2_24

• **[S:M4]** David I. Harvey, Stephen J. Leybourne and Paul Newbold (1997). *Testing the equality of prediction mean squared errors*. International Journal of Forecasting. https://doi.org/10.1016/s0169-2070(96)00719-4

• **[S:M9]** Antje Inness et al. (2019). *The CAMS reanalysis of atmospheric composition*. Atmospheric chemistry and physics. https://doi.org/10.5194/acp-19-3515-2019

• **[C2]** Xiaoye Jin et al. (2022). *Machine learning driven by environmental covariates to estimate high-resolution PM2.5 in data-poor regions*. PeerJ. https://doi.org/10.7717/peerj.13203

• **[S:M10]** Guolin Ke et al. (2017). *LightGBM: A Highly Efficient Gradient Boosting Decision Tree*. HAL (Le Centre pour la Communication Scientifique Directe).

• **[A8]** Jie Liu et al. (2025). *Characteristics of salt dust aerosols and their transport implications in the Aral Sea*. Scientific Reports. https://doi.org/10.1038/s41598-025-86880-5

• **[S:M11]** Scott Lundberg and Su■In Lee (2017). *A Unified Approach to Interpreting Model Predictions*. arXiv (Cornell University). https://doi.org/10.48550/arxiv.1705.07874

• **[C3]** Gideon Mazuruse et al. (2026). *Explainable PSO-optimised machine learning models for multi-pollutant air quality forecasting in major African cities with transfer learning*. Frontiers in Environmental Science. https://doi.org/10.3389/fenvs.2026.1828162

• **[S:M2]** Hanna Meyer et al. (2019). *Importance of spatial predictor variable selection in machine learning applications – Moving from data reproduction to spatial prediction*. Ecological Modelling. https://doi.org/10.1016/j.ecolmodel.2019.108815

• **[S:M5]** Whitney K. Newey and Kenneth D. West (1987). *A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix*. Econometrica. https://doi.org/10.2307/1913610

• **[C4]** Jianjun Ni et al. (2022). *An Improved Hybrid Transfer Learning-Based Deep Learning Model for PM2.5 Concentration Prediction*. Applied Sciences. https://doi.org/10.3390/app12073597

• **[A6]** Stefanos Papagiannis et al. (2024). *Air quality challenges in Central Asian urban areas: a PM2.5 source apportionment analysis in Dushanbe, Tajikistan*. Environmental Science and Pollution Research. https://doi.org/10.1007/s11356-024-33833-6

- **[S:M6]** Philipp Probst, Marvin N. Wright and Anne■Laure Boulesteix (2019). *Hyperparameters and tuning strategies for random forest*. Wiley Interdisciplinary Reviews Data Mining and Knowledge Discovery. https://doi.org/10.1002/widm.1301
- **[S:M1]** David R. Roberts et al. (2016). *Cross■validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure*. Ecography. https://doi.org/10.1111/ecog.02881
- **[B2]** Dié Tang, Yu Zhan and Fumo Yang (2024). *A review of machine learning for modeling air quality: Overlooked but important issues*. Atmospheric Research. https://doi.org/10.1016/j.atmosres.2024.107261
- **[A2]** Madina Tursumbayeva et al. (2023). *Cities of Central Asia: New hotspots of air pollution in the world*. Atmospheric Environment. https://doi.org/10.1016/j.atmosenv.2023.119901
- **[A5]** Kazbek Tursun et al. (2025). *Dominant sources of PM2.5 in Kazakhstan's urban cities: A PMF and HYSPLIT-based study for air quality management in Central Asia*. Urban Climate. https://doi.org/10.1016/j.uclim.2025.102706
- **[S:M7]** Yongming Xu et al. (2018). *Evaluation of machine learning techniques with multiple remote sensing datasets in estimating monthly concentrations of ground-level PM2.5*. Environmental Pollution. https://doi.org/10.1016/j.envpol.2018.08.029
- **[S:M12]** Yixuan Zheng et al. (2015). *Estimating ground-level PM2.5 concentrations over three megalopolises in China using satellite-derived aerosol optical depth measurements*. Atmospheric Environment. https://doi.org/10.1016/j.atmosenv.2015.06.046
- **[S:M8]** Tongshu Zheng et al. (2018). *Field evaluation of low-cost particulate matter sensors in high- and low-concentration environments*. Atmospheric measurement techniques. https://doi.org/10.5194/amt-11-4823-2018

Sources not indexed by the resolvers

- **[A1]** OpenAQ (2025). *Air Quality Data: Central Asia — regional snapshot of the 2024 Global Air Quality Data Landscape report*. OpenAQ (policy report). — read in full; not a journal article, so absent from Crossref/OpenAlex
- **[A3]** Guttikunda, S. (2024). *A multi-pollutant emissions inventory and air quality analysis for Bishkek*. Zenodo 12720883 (CC-BY). — read in full; Zenodo deposit, not indexed by the APIs used here

Sought but not obtained

Listed so the gap is visible rather than absent. No claim in this manuscript rests on these.

- **[A9]** query: *Particulate matter PM2.5 prediction in Tashkent using machine learning* — no matching record.